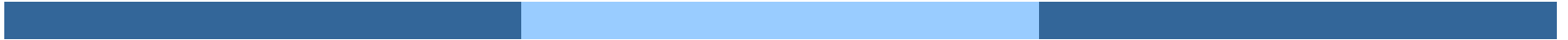
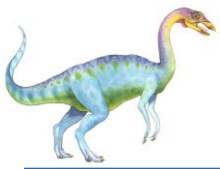


CPU Scheduling





Outline

- Basic Concepts
- Scheduling Criteria
- Scheduling Algorithms





Objectives

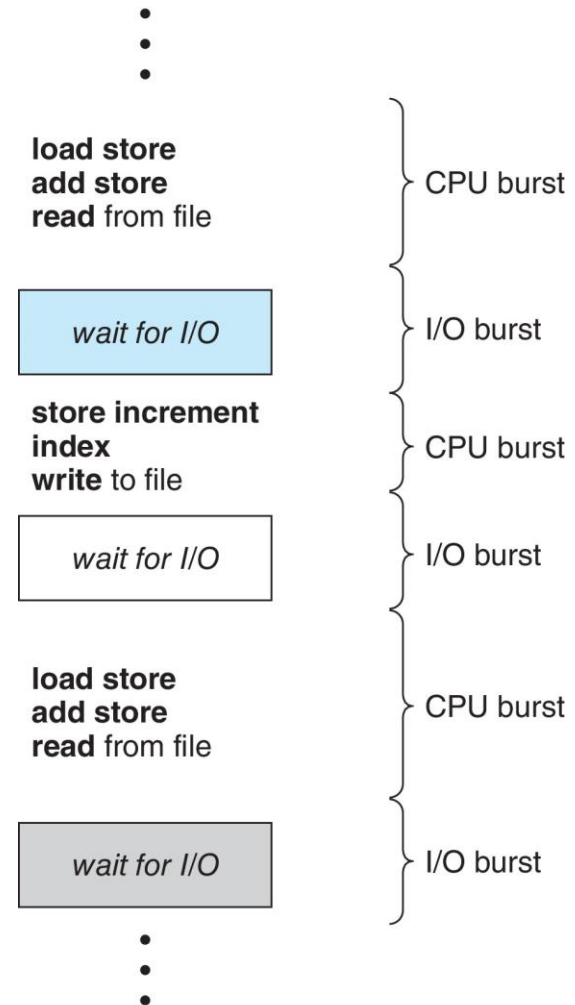
- Describe various CPU scheduling algorithms
- Assess CPU scheduling algorithms based on scheduling criteria
- Explain the issues related to multiprocessor and multicore scheduling
- Describe various real-time scheduling algorithms
- Describe the scheduling algorithms used in the Windows, Linux, and Solaris operating systems
- Apply modeling and simulations to evaluate CPU scheduling algorithms





Basic Concepts

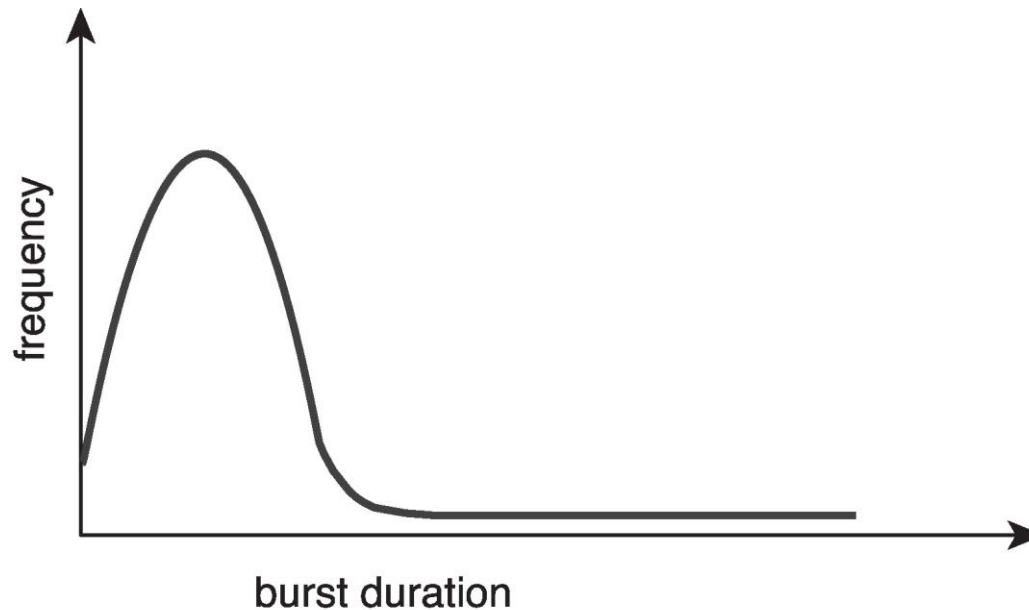
- Maximum CPU utilization obtained with multiprogramming
- CPU–I/O Burst Cycle – Process execution consists of a **cycle** of CPU execution and I/O wait
- **CPU burst** followed by **I/O burst**
- CPU burst distribution is of main concern





Histogram of CPU-burst Times

- Large number of short bursts
- Small number of longer bursts
- Histogram

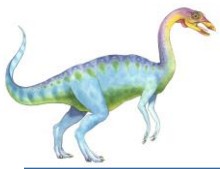




CPU Scheduler

- The **CPU scheduler** selects from among the processes in ready queue, and allocates a CPU core to one of them
 - The ready queue may be ordered in various ways
- CPU scheduling decisions may take place when a process:
 1. Switches from running to waiting state
 2. Switches from running to ready state
 3. Switches from waiting to ready
 4. Terminates
- For situations 1 and 4, there is no choice in terms of scheduling. A new process (if one exists in the ready queue) must be selected for execution.
- For situations 2 and 3, however, there is a choice.

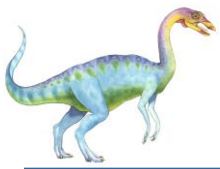




Preemptive and Nonpreemptive Scheduling

- When scheduling takes place only under circumstances 1 and 4, the scheduling scheme is **nonpreemptive**.
- Otherwise, it is **preemptive**.
- Under Nonpreemptive scheduling, once the CPU has been allocated to a process, the process keeps the CPU until it releases it either by terminating or by switching to the waiting state.
 - What is the potential problem?
- Virtually all modern operating systems including Windows, MacOS, Linux, and UNIX use preemptive scheduling algorithms.





Preemptive Scheduling and Race Conditions

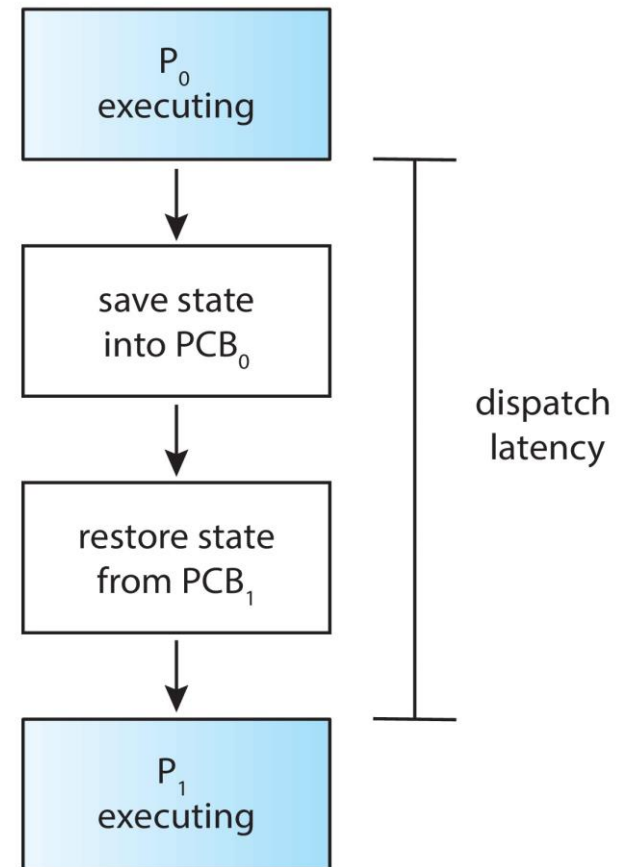
- Preemptive scheduling can result in race conditions when data are shared among several processes.
- Consider the case of two processes that share data. While one process is updating the data, it is preempted so that the second process can run. The second process then tries to read the data, which are in an inconsistent state.
 - We saw this in the bounded buffer example
- This issue will be explored in detail in Chapter 6.





Dispatcher

- Dispatcher module gives control of the CPU to the process selected by the CPU scheduler; this involves:
 - Switching context
 - Switching to user mode
 - Jumping to the proper location in the user program to restart that program
- **Dispatch latency** – time it takes for the dispatcher to stop one process and start another running

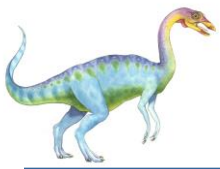




Scheduling Criteria

- **CPU utilization** – keep the CPU as busy as possible
- **Throughput** – # of processes that complete their execution per time unit
- **Turnaround time** – amount of time to execute a particular process
- **Waiting time** – amount of time a process has been waiting in the ready queue
- **Response time** – amount of time it takes from when a request was submitted until the first response is produced.





Optimization Criteria for Scheduling Algorithms

- Max CPU utilization
- Max throughput
- Min turnaround time
- Min waiting time
- Min response time





First- Come, First-Served (FCFS) Scheduling

- FCFS follows a non-preemptive approach, meaning, once the CPU lets a process take control, it won't preempt until the job terminates.
- It follows the criteria of arrival time for the selection process.
- The processor selects the very first job in the ready queue, and it runs till completion.
- It supports a pre-emptive and non-preemptive scheduling algorithm.
- All the jobs execute on a first-come, first-serve basis.
- Smaller processes take the lead in case of a tie.
- The general wait time is quite high because of the modus operandi that FCFS follows.
- The FCFS algorithm is better for the processes with comparatively large burst time since it involves no context switch between the processes.





First- Come, First-Served (FCFS) Scheduling

- Example with 3 processes

<u>Process</u>	<u>Burst Time</u>
P_1	24
P_2	3
P_3	3

- Suppose that the processes arrive in the order: P_1, P_2, P_3
The Gantt Chart for the above schedule is:



- Waiting time for $P_1 = 0$; $P_2 = 24$; $P_3 = 27$
- Average waiting time: $(0 + 24 + 27)/3 = 17$





FCFS Scheduling (Cont.)

Suppose that the processes arrive in the order:

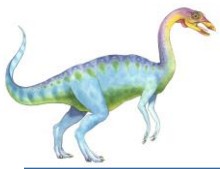
$$P_2, P_3, P_1$$

- The Gantt chart for the schedule is:



- Waiting time for $P_1 = 6$; $P_2 = 0$; $P_3 = 3$
- Average waiting time: $(6 + 0 + 3)/3 = 3$
- Much better than previous case
- **Convoy effect** - short process behind long process
 - Consider one CPU-bound and many I/O-bound processes





Shortest-Job-First (SJF) Scheduling

- Associate with each process the length of its next CPU burst
 - Use these lengths to schedule the process with the shortest time
- SJF is optimal – gives minimum average waiting time for a given set of processes
- One of the demerit of SJF is **starvation**.
 - **Starvation**: Longer jobs may suffer from starvation as they wait indefinitely behind short jobs, leading to potential delays in their execution.

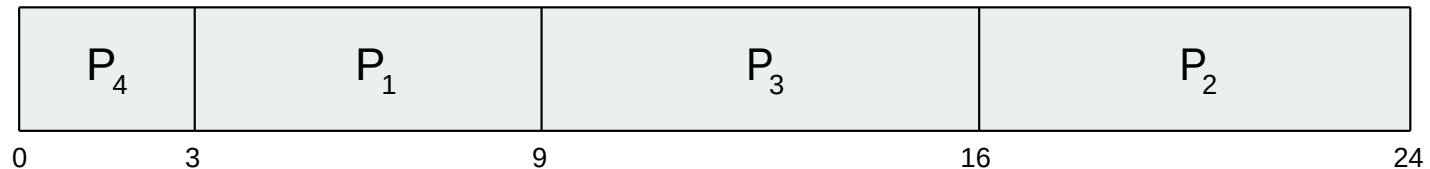




Example of SJF

<u>Process</u>	<u>Burst Time</u>
P_1	6
P_2	8
P_3	7
P_4	3

- SJF scheduling chart



- Average waiting time = $(3 + 16 + 9 + 0) / 4 = 7$





Determining Length of Next CPU Burst

- Can only estimate the length – should be similar to the previous one
 - Then pick process with shortest predicted next CPU burst
- Can be done by using the length of previous CPU bursts, using exponential averaging

1. t_n = actual length of n^{th} CPU burst
2. τ_{n+1} = predicted value for the next CPU burst
3. $\alpha, 0 \leq \alpha \leq 1$
4. Define :

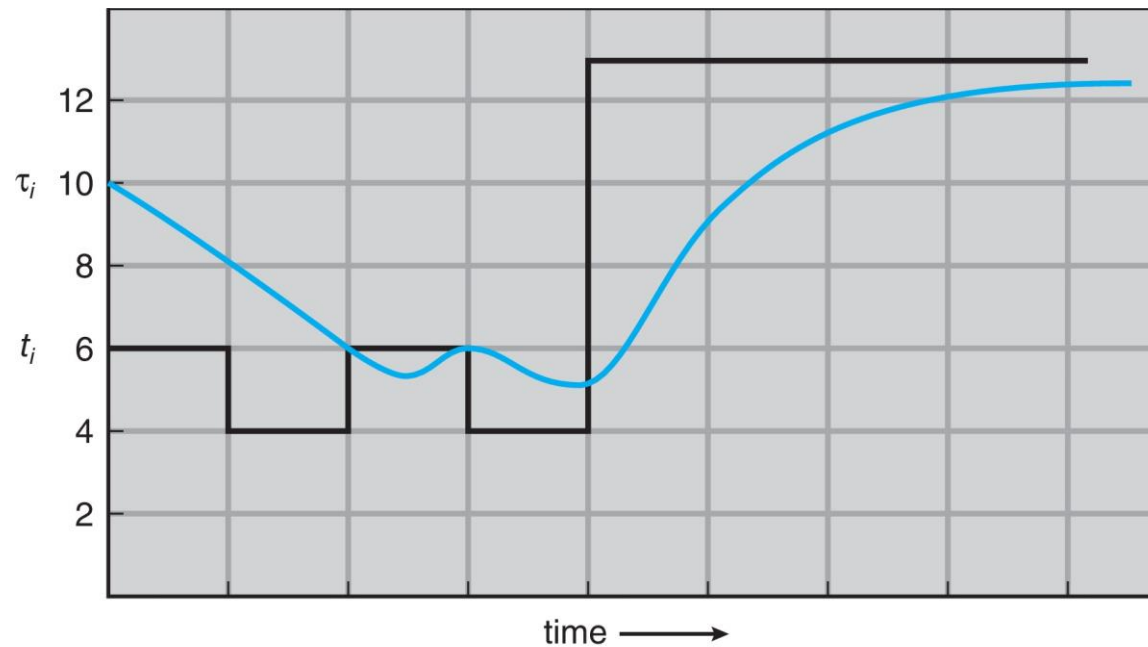
$$\tau_{n+1} = \alpha t_n + (1 - \alpha)\tau_n.$$

- Commonly, α set to $\frac{1}{2}$



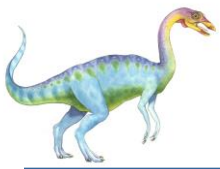


Prediction of the Length of the Next CPU Burst



CPU burst (t_i)	6	4	6	4	13	13	13	...
"guess" (τ_i)	10	8	6	5	9	11	12	...





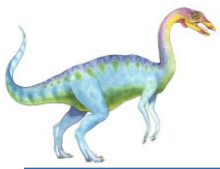
Examples of Exponential Averaging

- $\alpha = 0$
 - $\tau_{n+1} = \tau_n$
 - Recent history does not count
- $\alpha = 1$
 - $\tau_{n+1} = \alpha t_n$
 - Only the actual last CPU burst counts
- If we expand the formula, we get:

$$\begin{aligned}\tau_{n+1} = & \alpha t_n + (1 - \alpha)\alpha t_{n-1} + \dots \\ & + (1 - \alpha)^j \alpha t_{n-j} + \dots \\ & + (1 - \alpha)^{n+1} \tau_0\end{aligned}$$

- Since both α and $(1 - \alpha)$ are less than or equal to 1, each successive term has less weight than its predecessor





Shortest Remaining Time First Scheduling

- Preemptive version of SJN
- Whenever a new process arrives in the ready queue, the decision on which process to schedule next is redone using the SJN algorithm.
- Is SRT more “optimal” than SJN in terms of the minimum average waiting time for a given set of processes?



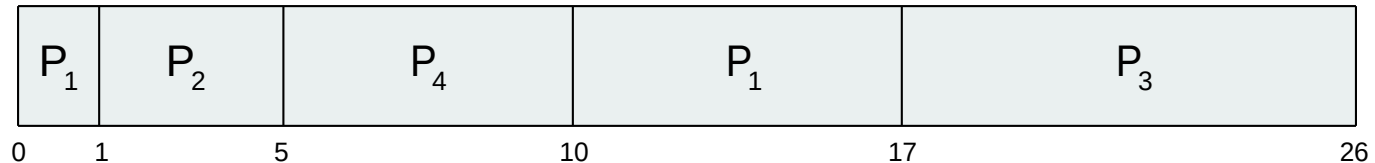


Example of Shortest-remaining-time-first

- Now we add the concepts of varying arrival times and preemption to the analysis

<u>Process</u>	<u>Arrival Time</u>	<u>Burst Time</u>
P_1	0	8
P_2	1	4
P_3	2	9
P_4	3	5

- Preemptive* SJF Gantt Chart



- Average waiting time = $[(10-1)+(1-1)+(17-2)+(5-3)]/4 = 26/4 = 6.5$





Round Robin (RR)

- The round-robin (RR) scheduling algorithm is designed especially for **time-sharing systems**. It is similar to FCFS scheduling, but pre-emption is added to switch between processes.
- A small unit of time, called a **time quantum** or **time slice**. A time quantum is generally from 10 to 100 milliseconds. The ready queue is treated as a circular queue.
- Each process gets a small unit of CPU time (**time quantum q**). After this time has elapsed, the process is preempted and added to the end of the ready queue.
- If there are n processes in the ready queue and the time quantum is q , then each process gets $1/n$ of the CPU time in chunks of at most q time units at once. No process waits more than $(n-1)q$ time units.
- Timer interrupts every quantum to schedule next process
- Performance
 - q large $\Rightarrow$ FIFO
 - q small $\Rightarrow q$ must be large with respect to context switch, otherwise overhead is too high

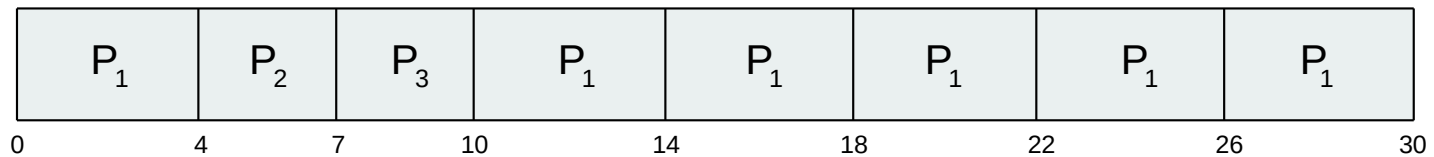




Example of RR with Time Quantum = 4

<u>Process</u>	<u>Burst Time</u>
P_1	24
P_2	3
P_3	3

- The Gantt chart is:

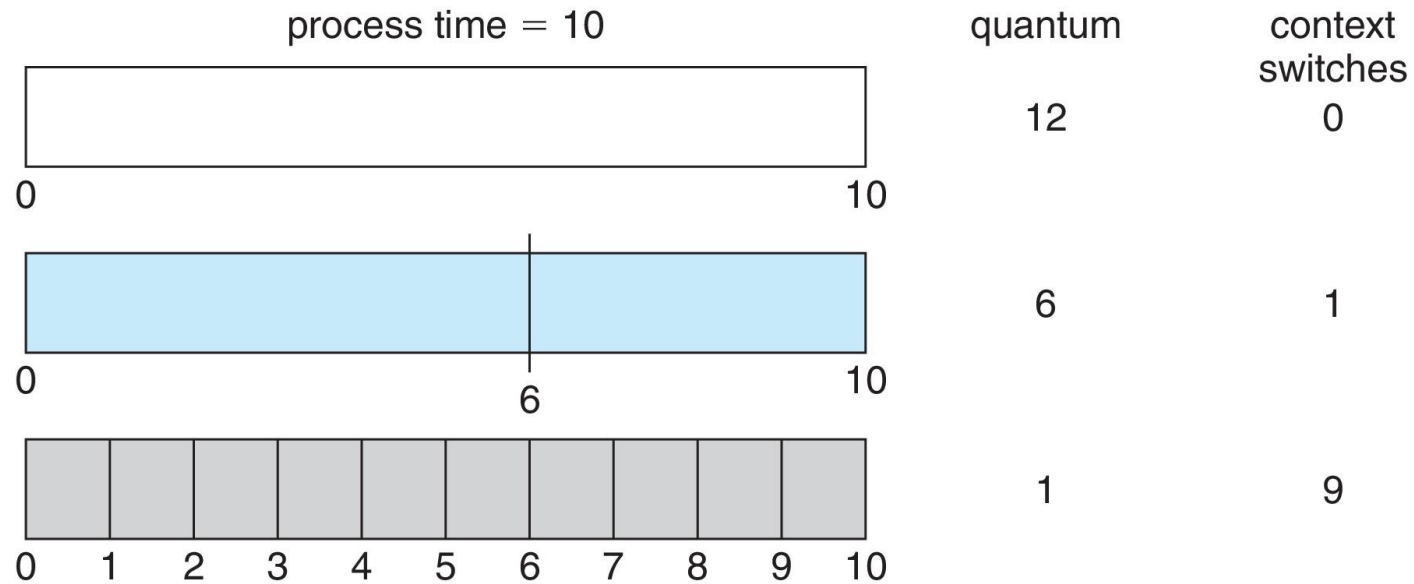


- Typically, higher average turnaround than SJF, but better **response**
- q should be large compared to context switch time
 - q usually 10 milliseconds to 100 milliseconds,
 - Context switch < 10 microseconds



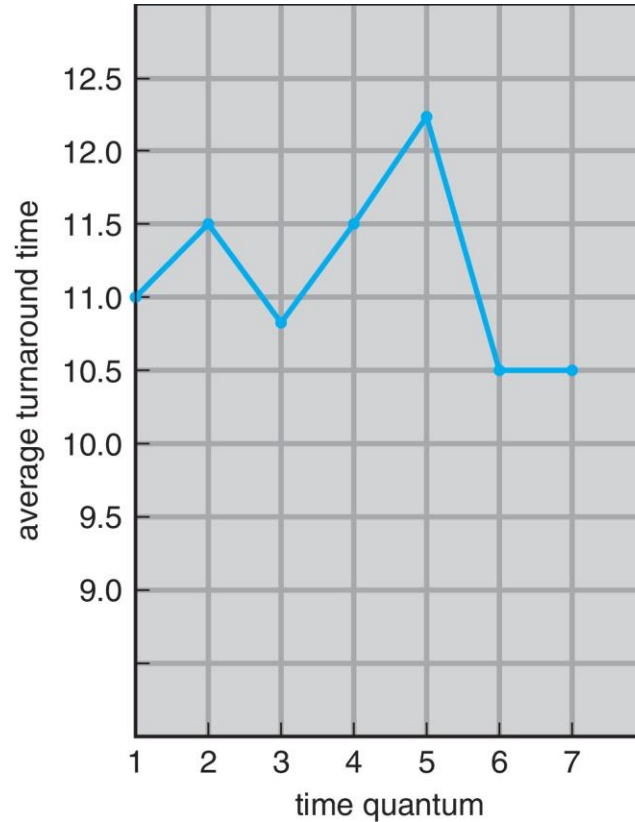


Time Quantum and Context Switch Time





Turnaround Time Varies With The Time Quantum



process	time
P_1	6
P_2	3
P_3	1
P_4	7

80% of CPU bursts
should be shorter than q





Examples

1. Round robin scheduling falls under the category of _____
 - a) Non-preemptive scheduling
 - b) Preemptive scheduling
 - c) All of the mentioned
 - d) None of the mentioned

2. With round robin scheduling algorithm in a time shared system _____
 - a) using very large time slices converts it into First come First served scheduling algorithm
 - b) using very small time slices converts it into First come First served scheduling algorithm
 - c) using extremely small time slices increases performance
 - d) using very small time slices converts it into Shortest Job First algorithm





Priority Scheduling

- A priority number (integer) is associated with each process
- The CPU is allocated to the process with the highest priority (usually, smallest integer $\equiv$ highest priority)
- Two schemes:
 - Preemptive
 - Nonpreemptive
- Problem $\equiv$ **Starvation** – low priority processes may never execute
- Solution $\equiv$ **Aging** – as time progresses increase the priority of the process
- Note: SJF is priority scheduling where priority is the inverse of predicted next CPU burst time





Example of Priority Scheduling

<u>Process</u>	<u>Burst Time</u>	<u>Priority</u>
P_1	10	3
P_2	1	1
P_3	2	4
P_4	1	5
P_5	5	2

- Priority scheduling Gantt Chart



- Average waiting time = 8.2





Example: GATE 2017

Consider the set of processes with arrival time (in milliseconds), CPU burst time (in milliseconds), and priority (0 is the highest priority) shown below. None of the processes have I/O burst time.

Process	Arrival Time	Burst Time	Priority
P_1	0	11	2
P_2	5	28	0
P_3	12	2	3
P_4	2	10	1
P_5	9	16	4

The average waiting time (in milliseconds) of all the processes using preemptive priority scheduling algorithm is ____.

Note: This question appeared as Numerical Answer Type.

- (A) 29
- (B) 30
- (C) 31
- (D) 32



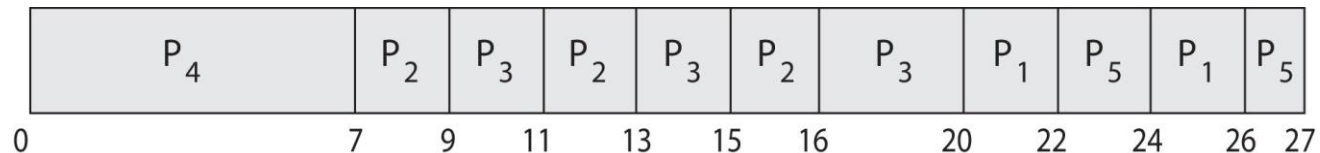


Priority Scheduling w/ Round-Robin

- Run the process with the highest priority. Processes with the same priority run round-robin
- Example:

<u>Process</u>	<u>Burst Time</u>	<u>Priority</u>
P_1	4	3
P_2	5	2
P_3	8	2
P_4	7	1
P_5	3	3

- Gantt Chart with time quantum = 2





MCQ

Q1. Which of the following statements are true? (GATE 2010)

- I. Shortest remaining time first scheduling may cause starvation
 - II. Preemptive scheduling may cause starvation
 - III. Round robin is better than FCFS in terms of response time
- a) only
 - b) I and III only
 - c) II and III only
 - d) I, II and III

Q2. Which of the following process scheduling algorithm may lead to starvation

- a) FIFO
- b) Round Robin
- c) Shortest Job Next
- d) None

Q3. Which of the following scheduling algorithms may cause starvation ?

- a. First-come-first-served
 - b. Round Robin
 - c. Priority
 - d. Shortest process next
 - e. Shortest remaining time first
- (A) a, c and e
 - (B) c, d and e
 - (C) b, d and e
 - (D) b, c and d





Example:1

Consider the 3 processes, P1, P2 and P3 shown in the table.

Process	Arrival time	Time Units Required
P1	0	5
P2	1	7
P3	3	4

The completion order of the 3 processes under the policies FCFS and RR2 (round robin scheduling with CPU quantum of 2 time units) are

(A)

FCFS: P1, P2, P3
RR2: P1, P2, P3

(B)

FCFS: P1, P3, P2
RR2: P1, P3, P2

(C)

FCFS: P1, P2, P3
RR2: P1, P3, P2

(D)

FCFS: P1, P3, P2
RR2: P1, P2, P3





Example:2

An operating system uses shortest remaining time first scheduling algorithm for pre-emptive scheduling of processes. Consider the following set of processes with their arrival times and CPU burst times (in milliseconds):

Process	Arrival Time	Burst Time
P1	0	12
P2	2	4
P3	3	6
P4	8	5

The average waiting time (in milliseconds) of the processes is _____.

- (A) 4.5
- (B) 5.0
- (C) 5.5
- (D) 6.5





Example:3

Consider the following CPU processes with arrival times (in milliseconds) and length of CPU bursts (in milliseconds) except for process P4 as given below:

Process	Arrival Time	Burst Time
P1	0	5
P2	1	1
P3	3	3
P4	4	x

If the average waiting time across all processes is 2 milliseconds and pre-emptive shortest remaining time first scheduling algorithm is used to schedule the processes, then find the value of x ?

- (A) 1
- (B) 2
- (C) 4
- (D) None of these





Example:4

Four jobs to be executed on a single processor system arrive at time 0 in the order A, B, C, D. Their burst CPU time requirements are 4, 1, 8, 1 time units respectively. The completion time of A under round robin scheduling with time slice of one time unit is

- (A) 10
- (B) 4
- (C) 8
- (D) 9





Example-5

Q: Consider three CPU-intensive processes, which require 10, 20 and 30 time units and arrive at times 0, 2 and 6, respectively. How many context switches are needed if the operating system implements a shortest remaining time first scheduling algorithm? Do not count the context switches at time zero and at the end.

- a) 1
- b) 2
- c) 3
- d) 4





Example

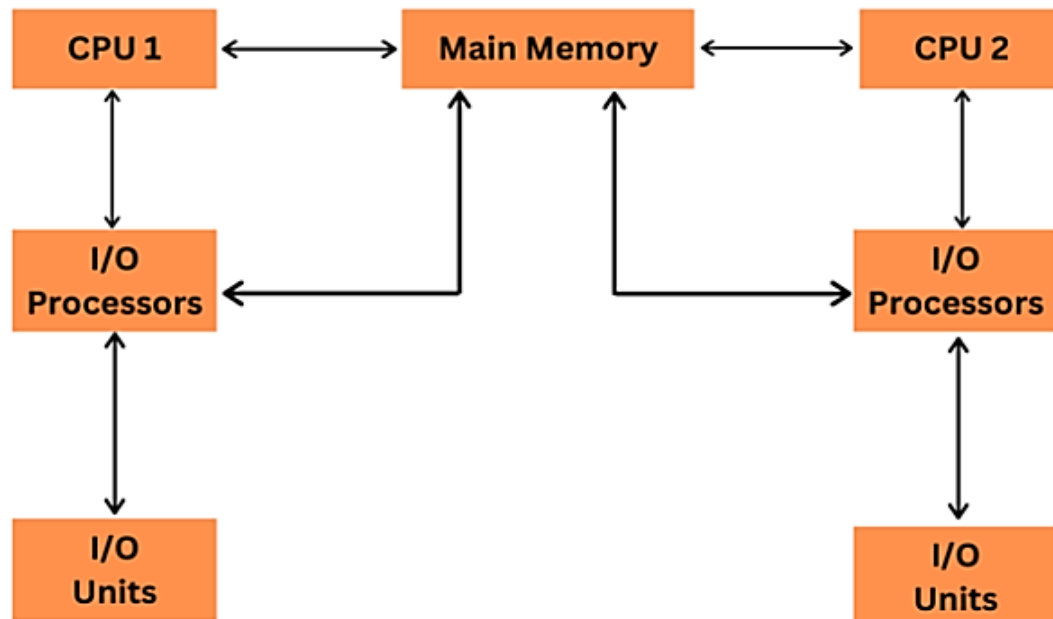
Consider the set of four processes P1, P2, P3, and P4, each with a process execution time of 5, 10, 15, and 20 milliseconds. The process execution consists of 80% CPU burst time and 20% I/O burst time. If the CPU scheduling policy is round robin with time stamp 2 milliseconds. What will be the average turnaround time.





Multiprocessor scheduling

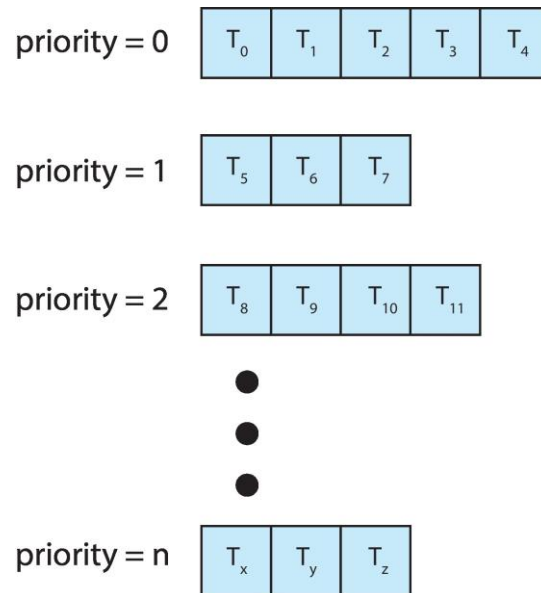
- **Multiple processor scheduling** or multiprocessor scheduling focuses on designing the system's scheduling function, which consists of more than one processor.
- To increase the system's overall performance, numerous processors or cores are frequently used in modern computer systems.
- Multiple processor scheduling involves deciding which processes should be assigned to which processor or core and how long they should be permitted to run.
- While ensuring that all processes are fairly and appropriately prioritized, the objective is to achieve efficient utilization of the available processors.





Multilevel Queue

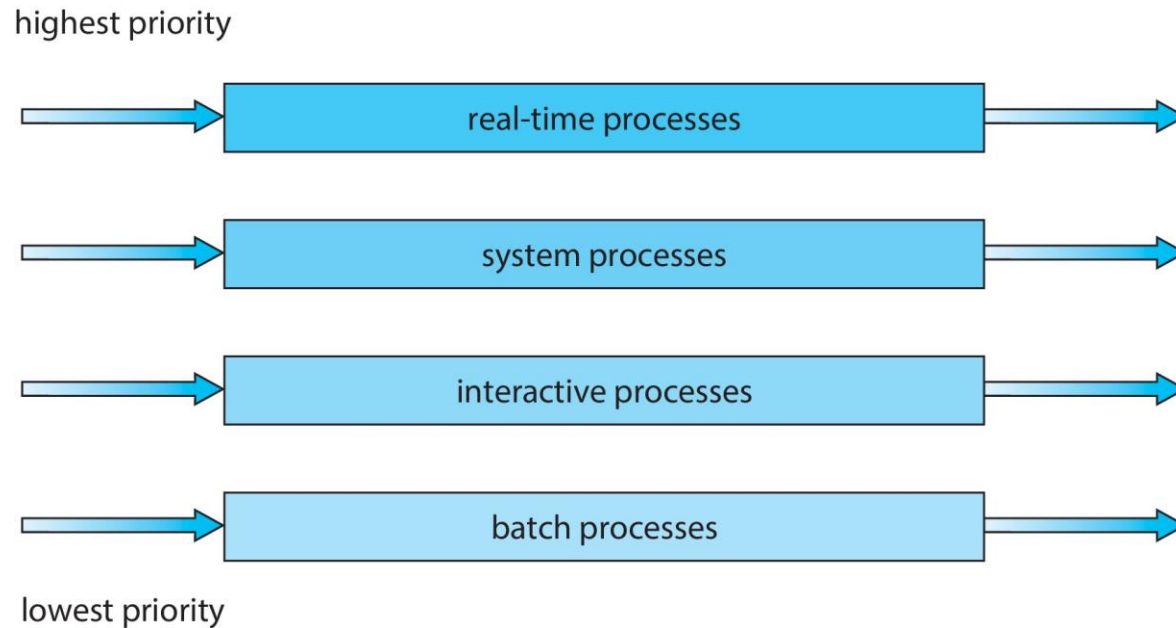
- The ready queue consists of multiple queues
- Example:
 - Priority scheduling, where each priority has its separate queue.
 - Schedule the process in the highest-priority queue!





Multilevel Queue

- Prioritization based upon process type





Multilevel Feedback Queue

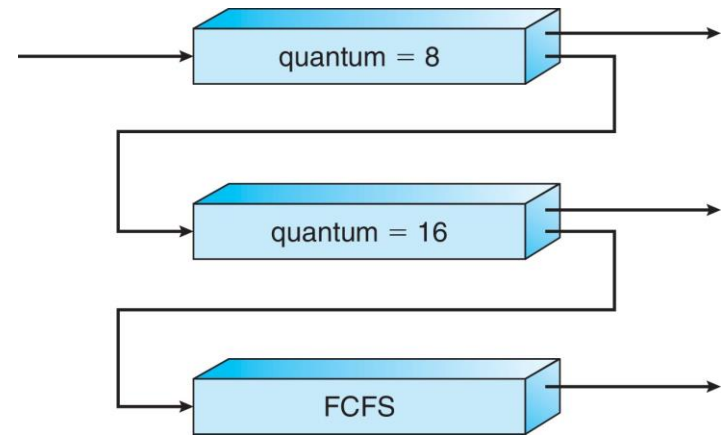
- A process can move between the various queues.
- Multilevel-feedback-queue scheduler defined by the following parameters:
 - Number of queues
 - Scheduling algorithms for each queue
 - Method used to determine when to upgrade a process
 - Method used to determine when to demote a process
 - Method used to determine which queue a process will enter when that process needs service
- Aging can be implemented using multilevel feedback queue





Example of Multilevel Feedback Queue

- Three queues:
 - Q_0 – RR with time quantum 8 milliseconds
 - Q_1 – RR time quantum 16 milliseconds
 - Q_2 – FCFS
- Scheduling
 - A new process enters queue Q_0 which is served in RR
 - ▶ When it gains CPU, the process receives 8 milliseconds
 - ▶ If it does not finish in 8 milliseconds, the process is moved to queue Q_1
 - At Q_1 job is again served in RR and receives 16 additional milliseconds
 - ▶ If it still does not complete, it is preempted and moved to queue Q_2



End

